

Design of control strategy for reactor power control system of LBE cooled Actinide Burner Reactor (ABR) using system analysis code NUSOL-LMR

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ABSTRACT

The power control strategy for the Lead–Bismuth Eutectic (LBE) cooled Actinide Burner Reactor (ABR) has been designed to assess the closed-loop performance of newly developed control blocks that were not modeled in the NUSOL-LMR code before this research effort. After establishing a sustainable model of ABR in NUSOL-LMR, validation of the model in light of the already-published research is performed. The said closed-loop control strategy (consisting of the power and temperature control blocks) is evaluated under different transient scenarios (reactivity insertion, primary coolant pump fault, and change in demand power levels). The power and temperature control blocks concurrently control reactor power during any transient occurrence. The coupling effect of the primary coolant temperature deviation on the reactor power has been minimized by the proportional and integral (PI) controller in the temperature control block. The simulation results demonstrate that the closed-loop performance of the ABR under the controller action during transient events is quite satisfactory. The closed-loop power regulation performance of the conventional sliding mode controller (SMC) with a PI sliding surface (PISM) in the power control block of NUSOL-LMR is better than that of a simple proportional, integral and derivative (PID) controller. The asymptotic stability of the closed-loop system with the PISM controller has been ensured using the Lyapunov stability theory.

1. Introduction

The lead-cooled fast reactor (LFR) uses lead or LBE as a coolant. It is one of the six auspicious nuclear energy technologies chosen by the Generation IV International Forum in 2002. In addition, the LFR offers much potential for achieving the objectives outlined by the forum (Kelly, 2014; Pioro, 2016; Cinotti et al., 2011). Research and development on LFR's technology and design is being conducted in various parts of the globe, including Europe, USA, China, Russia, and others. The several LFR types are listed in Table 1.

The ABR is one of the types of LFR in which the LBE is used as a primary coolant. Initially, safety analysis on the preliminary design of the ABR was carried out by considering the natural circulation of the LBE (Davis et al., 2002). Later, the ABR's preliminary design was modified, and the altered design under the forced circulation of the LBE was evaluated under different transient conditions (Davis, 2003).

Controlling a nuclear reactor and all of its associated subsystems is quite challenging. Safety is one of the most significant concerns when discussing nuclear technology. Therefore, it is an utmost requirement to devise a nuclear reactor's control strategy for its efficient and trustworthy operation. Thus, it makes the reactor power control system

particularly important as it regulates the power by automatically regulating the movement of control rods under different transient situations. In addition, the control schemes of nuclear reactor not only make certain that the reactor is safe but also contribute to reducing financial constraints by avoiding the activation of the reactor protection system. Hence, keeping the reactor in operation without any power outages.

Pressurized water reactor (PWR) control systems have decades of operational experience and are more established than those of LFRs. High core temperatures and a sizable coolant temperature differential between the inlet and outlet of an LFR's core are two distinguishing features that set apart the power control strategies of LFRs from those of PWRs. With the passage of time, research is being carried out on the control schemes of various types of LFRs, as discussed in Table 1. By using the feedforward-feedback control approach, the fan blower speed is regulated to maintain a constant coolant temperature in the secondary loop of the Italian LBE-XADS (Cammi et al., 2006). Likewise, the Chinese design (CLEAR-1B) has employed the same feedforward-feedback control technique with just a minor modification in the control structure; it also has an additional power controller (PID), which adjusts the control rod movement to keep the power level close to the reference

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Nomenclature

Symbols

A	The cross-sectional area (m^2)
C_p	The specific heat at constant pressure (kJ/kg K)
c_i	Precursor concentration in i th group ($1/\text{m}^3$)
c_{ri}	Normalized precursor concentration in i th group
e	Error
$\dot{e}$	Derivative of error
F_w	Wall frictional Force (N/m)
G_r	Control rod worth ($1/\text{m}$)
H	Specific enthalpy (J/kg)
k_d	Derivative gain of PID
k_i	Integral Gain of PID
k_p	Proportional gain of PID
k_1	Proportional gain in PISM
k_2	Integral gain in PISM
k_3	Reaching law gain in PISM
n	Density of neutrons ($1/\text{m}^3$)
n_r	Normalized neutron density
P_{ref}	Reference power (MW)
P_{out}	Output power (MW)
$\dot{P}_{ref}$	Derivative of reference power (MW)
$\dot{P}_{out}$	Derivative of output power (MW)
Q_w	Wall-to-flow heat transfer per volume (W/m^3)
S	Source term of heat conduction equation (W/m^3)
s	Sliding surface
$\dot{s}$	Derivative of sliding surface
T	Temperature of heat structure node (K)
T_c	Coolant average temperature (K)
T_{co}	Initial coolant average temperature (K)
T_f	Fuel average temperature (K)
T_{fo}	Initial fuel average temperature (K)
V	Flow velocity (m/s)
Z	Control volume length (m)
Z_r	Control rod speed (m/s)
α_c	Coolant temperature feedback coefficient
α_d	Coolant density feedback coefficient
ρ_c	Coolant average density (kg/m^3)
κ	Coefficient of thermal conductivity (W/m K)
ρ	Density (kg/m^3)
β_i	Delayed neutron fraction of i th group
λ_i	Decay constants of precursors of i th group ($1/\text{s}$)
ρ_e	External step reactivity disturbance ($\$$)

ρ_f	Feedback reactivity ($\$$)
α_f	Fuel temperature feedback coefficient
ρ_{c0}	Initial coolant average density (kg/m^3)
Λ	Prompt Neutron generation time (s)
ρ_r	Reactivity due to control rods ($\$$)
ω	Reaching law gain of PISM
ρ_t	Total reactivity ($\$$)
β	Total delayed neutron fraction

scheme for the M2LFR has recently been devised that combines the simultaneous operation of the power and temperature controllers to regulate the reactor's power. In addition, the PID controller gain settings have been optimized by using the fuzzy logic in this control approach (Shen et al., 2023).

The significance and importance of conventional and classical PI and PID controllers have always been predominant in the nuclear industry. These controllers require only a few parameters (k_p , k_i and k_d) to be tuned as compared to other complex control schemes and methodologies. However, the PID and PI based control algorithms also have certain limitations; they cannot show robustness against parameter variations, and uncertainties in the dynamical system.

The sliding mode control is a robust control design technique that can cope with the issues faced by the PID based control algorithm. SMC ensures the stability of the closed-loop system, despite the dynamical system being subjected to external disturbances and uncertainties. The SMC will force the system states to slide along the predefined sliding surface, making the closed-loop system response less sensitive to parameter variations and external disturbances. Owing to these benefits, SMCs have been widely used in the control strategies of nuclear reactors. Various SMC-based control schemes for PWRs (Hui and Yuan, 2021, 2020), high-temperature gas-cooled reactors (HTGR's) (Wang et al., 2019a,b), research reactors (Qaiser et al., 2009) etc., can be found in the literature. Despite having many advantages, a conventional SMC has one major chattering issue due to a discontinuous control signal in its control law. However, many chattering suppression methods are available in the literature to resolve this issue (Suleiman et al., 2018; Lee et al., 2009).

Control blocks containing different types of controllers are considered an integral part of any system analysis code for a nuclear reactor. Before this research work, these control blocks were not modeled in the NUSOL-LMR code. Therefore, the code lacked the capability to perform the closed-loop analysis of metal-cooled fast reactors under the controller action during transient events. In this regard, the main objective of the research work was to modify the code subroutines and develop the control blocks consisting of conventional PI, PID, and SMC in the code.

Moreover, after modification of the code, the next aim was to design the power control strategy for LFR to test the functionality of the developed control blocks. In this context, the ABR was used as a reference reactor and was modeled using the NUSOL-LMR code. In addition, the control strategy for power regulation of the ABR had not been designed in the literature before this research work. Therefore, after the model validation of the ABR using NUSOL-LMR, the power control strategy is devised with the concurrent working of power and temperature control blocks. The power control block is designed with an option to use either a classical PID controller or a slightly modified SMC with PI as a sliding surface (PISM). The power control block will regulate the output power of the ABR by maneuvering the speed of the control rods.

Furthermore, to avoid the effect of LBE temperature fluctuations on the reactor power, a simple PI temperature controller in the temperature control block is employed to control the mass flow rate of the secondary coolant (water). The PI controller will keep the LBE

value (Yan et al., 2014). With the aid of a PID controller, a control strategy for the BREST-300 reactor power control system has been developed, and the Laurent expansion technique has been used to determine the PID coefficients (Zarei et al., 2017). Additionally, a multi-PI controller-based decentralized control system has been designed for the ALFRED reactor's power regulation (Ponciroli et al., 2015). For the SNCLFR-100 (Duan et al., 2021), a power and temperature controller-based control technique has been adopted. A more effective control

Table 1
Different types of LFRs.

Origin	LFR type	Remarks
Europe	- ELFR (European Lead Fast Reactor) - ALFRED (Advanced Lead-cooled Fast Reactor European Demonstrator) - MYRRHA (Multi-purpose hYbrid Research Reactor for high-tech Applications) - SEALER (SwEdish Advanced LEad Reactor) - LBE-XADS (LBE-eXperimental Accelerator Driven System)	Bubelis et al. (2015) Lodi and Grasso (2018), Narcisi et al. (2020) Toti et al. (2018), Koloszar et al. (2020) Acharya et al. (2021), Hernandez et al. (2020) Cammi et al. (2006)
Russia	- SVBR-10/75/100 (Svintsovo Vismutovyi Bystryi Reaktor) - BREST-300/1200 (Bystryi Reaktor so Svintsovyim Teplonositelem)	Guo et al. (2015) Mosunova et al. (2020), Chudinova and Nikonov (2020)
USA	- STAR-LM (Secure Transportable Autonomous Reactor - Liquid Metal) - STAR-H2 (Secure Transportable Autonomous Reactor - Hydrogen production) - SUPERSTAR (SUstainable Proliferation-resistance Enhanced Refined Secure Transportable Autonomous Reactor) - SSTAR (Small Secure Transportable Autonomous Reactor) - ABR (Actinide Burner Reactor) - DLFR (Westinghouse Demonstration Lead Fast Reactor)	Sienicki and Petkov (2002) Bortot et al. (2011) Sienicki et al. (2007) Davis et al. (2002), Davis (2003), Mac Donald et al. (2001) Ferroni et al. (2017)
China	- CLEAR (China LEAD-based Reactor) - SNCLFR-100 (Small modular Natural Circulation Lead Cooled Fast Reactor) - M2LFR-1000 (Medium-modular Lead Cooled Fast Reactor)	Arthur et al. (2022) Li et al. (2017) Shen et al. (2019)
Japan	- PBWFR (Pb-Bi-cooled direct contact boiling Water Fast Reactor) - CANDLE (Constant Axial shape of Neutron flux, nuclide densities and power shape During Life of Energy producing reactor)	Tain et al. (2012) Khotyayintsev et al. (2020)
Korea	- PASCAR (Proliferation resistant, Accident-tolerant, Self-supported, Capsular and, Assured Reactor) - URANUS (Ubiquitous, Rugged, Accident-forgiving, Non-proliferating and Ultra-Lasting Sustainer)	Choi et al. (2011) Zhao et al. (2020)

temperature at the core intake near to its reference value during transient operating conditions.

The remaining part of the article is structured in the following sequence. The ABR nodalization scheme using NUSOL-LMR and model validation are discussed in Sections 2 and 3, respectively. In Section 4, the ABR reactor power control system is discussed in detail. Furthermore, the simulation and results are included in Section 5, and the research's conclusion is documented in Section 6.

2. The ABR nodalization scheme using NUSOL-LMR

The NUSOL-LMR is a fully implicit scheme-based system analysis code that can perform safety analysis of metal-cooled fast reactors during their power operation. The code uses a one-dimensional single-phase flow model to perform the calculations on the hydraulic components. The three basic governing equations of mass, energy and momentum for general-purpose one-dimensional hydraulic component are given below:

- Mass Conservation Equation

$$A \frac{\partial \rho}{\partial t} + A \frac{\partial \rho V}{\partial Z} = 0 \quad (1)$$

- Energy Conservation Equation

$$A \frac{\partial \rho H + (V^2/2)}{\partial t} + A \frac{\partial \rho V (H + V^2/2)}{\partial Z} = A \frac{\partial \rho}{\partial t} + AV \frac{\partial P}{\partial Z} + AV F_w + AQ_w \quad (2)$$

- Momentum Conservation Equation

$$A \rho \frac{\partial V}{\partial t} + A \rho V \frac{\partial V}{\partial Z} = -A \frac{\partial P}{\partial Z} - A \rho g \sin \theta - AF_w \quad (3)$$

Similarly, to calculate heat transfer from solid boundaries (such as the core, fuel rods, etc.) to the different kinds of fluids in the fast reactor system, the code employs a one-dimensional unsteady heat conduction equation to establish a thermal component heat structure model. The heat conduction equation is given as follows:

$$\rho C_p \frac{\partial T}{\partial t} = \kappa \left(\frac{1}{r} \frac{\partial T}{\partial r} + \frac{\partial^2 T}{\partial r^2} \right) + S \quad (4)$$

A detailed description of the equations mentioned above, different closure models, boundary conditions, and Newton-based fully implicit numerical scheme to solve these equations can be found in the literature (Qingling et al., 2022; Nourgaliev and Christon, 2012).

2.1. NUSOL-LMR ABR model

The comprehensive explanation of ABR's design and its working principle is described and documented in the literature (Davis, 2003; Mac Donald et al., 2001). In addition, the primary coolant system in which the LBE flows as the main coolant and the reactor vessel auxiliary cooling system (RVACS) of the ABR (Boardman et al., 2000) are shown in Fig. 1.

Each of the 157 fuel assemblies that make up the entire core of the ABR consists of 312 fuel pins (Hejzlar et al., 2002). To avoid the issue of reduction in reactivity worth caused by excessive positive coolant voiding, each fuel assembly incorporates a unique gas-filled streaming channel, both at the periphery and at the center. Fig. 2 demonstrates the fuel assembly with the streaming channels. The control rods are inserted in the central streaming channels of the designated fuel assemblies as per requirement.

The nodalization scheme of the ABR is developed using the NUSOL-LMR code and is shown in Fig. 3. The geometry of the ABR from which a model is derived is tabulated in Table 2, while Table 3 contains the specifics of every component in a model.

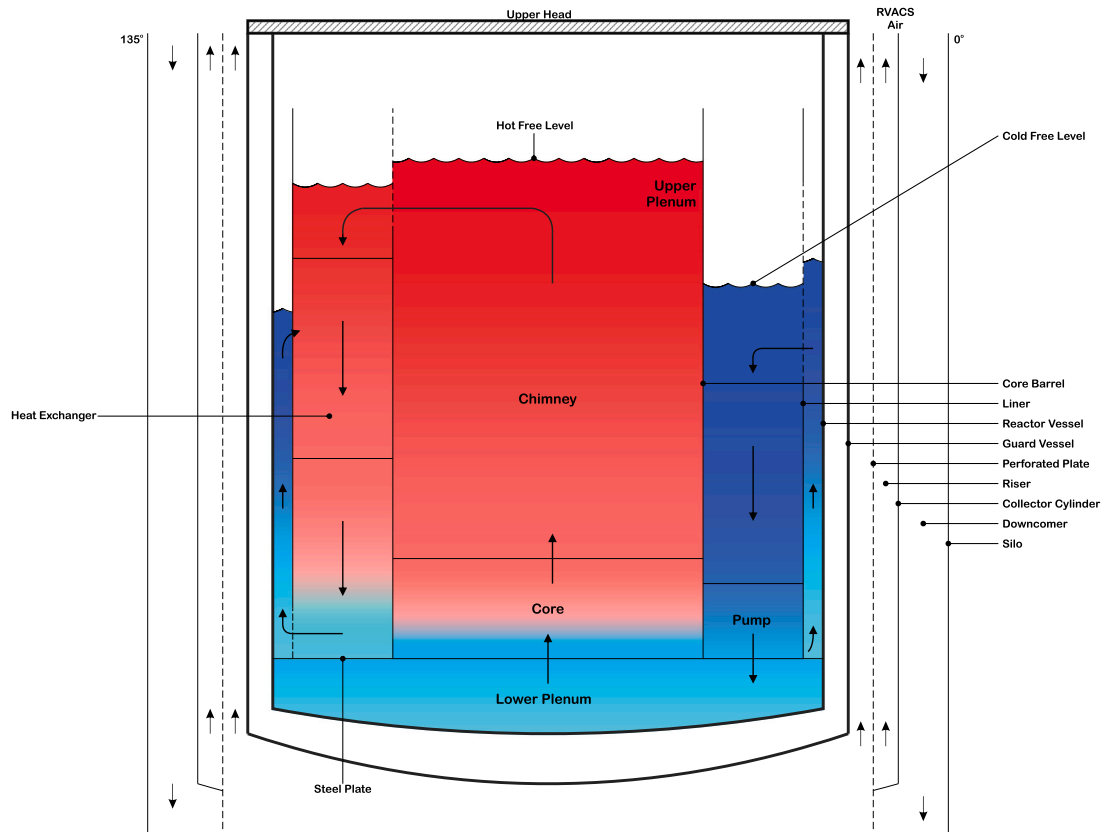


Fig. 1. Layout of the Actinide-Burner Reactor (ABR) (Mac Donald et al., 2001).

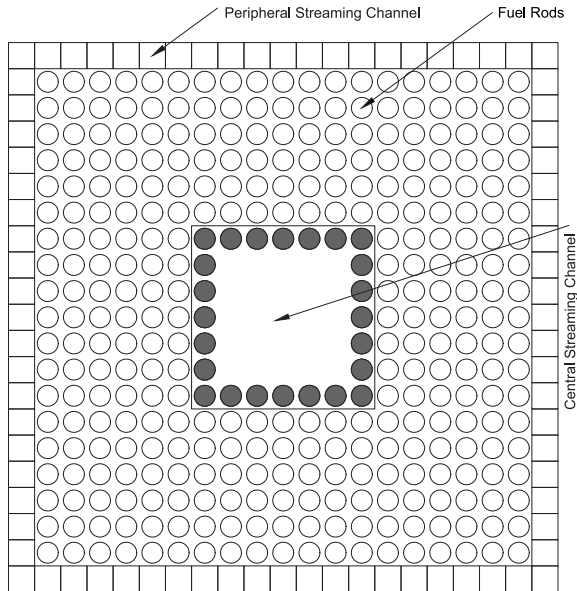


Fig. 2. Layout of a single fuel assembly of the ABR (Hejzlar et al., 2002).

The main coolant (LBE) draws the heat away from the core. After passing via the upper plenum area, LBE flows into the heat exchanger. The heat exchanger in the NUSOL-LMR code consists of a pair of pipes with a heat structure in between them, where the heat is exchanged from the main coolant (hot side) to the secondary coolant (cold side). Furthermore, the RVACS is modeled with the help of an air heat exchanger, where the air is circulating to take away heat from the main coolant flowing through the 'RISER' component (the annular area

located between the liner and reactor vessel). The main centrifugal pump placed in the primary loop forces the LBE to the core section after passing through the 'LOWER PLENUM' component. The addition of the 'BYPASS' component provides an alternate path for the LBE, skipping the core region and going directly towards the 'CHIMNEY' component.

Placing the time-dependent volume component (TDV1) at the very top of the loop maintains the pressure boundary of the entire primary loop. Similarly, ' $2^{nd}_{TDV_IN}$ ' and ' $2^{nd}_{TDV_OUT}$ ' are the time-dependent volume components that maintain the temperature and pressure boundary of the secondary loop, respectively. The entrance velocity of the secondary coolant (water) is managed via a time-dependent junction (TDJ). In the NUSOL-LMR model of the ABR shown in Fig. 3, the red highlighted sections represent the 'Heat Structures' components for thermal calculations.

3. Model validation

3.1. Steady-state calculations

The steady-state calculations for the ABR at a normal power level were carried out using the NUSOL-LMR model. Table 4 compares steady-state results from the NUSOL-LMR code with already published simulated results in the literature (Davis, 2003). This comparison demonstrates that the outcomes of the NUSOL-LMR and the ATHENA models are in excellent accord.

3.2. Transient-state calculations

After the steady-state calculations, the NUSOL-LMR code was used to perform the transient-state calculations on the ABR model, taking into account the unprotected overpower transient (UTOP). With the identical starting conditions as in Table 4, at 10 s, a step reactivity of 0.2\$ is inserted into the core. Figs. 4 (a) and 4 (b) demonstrate

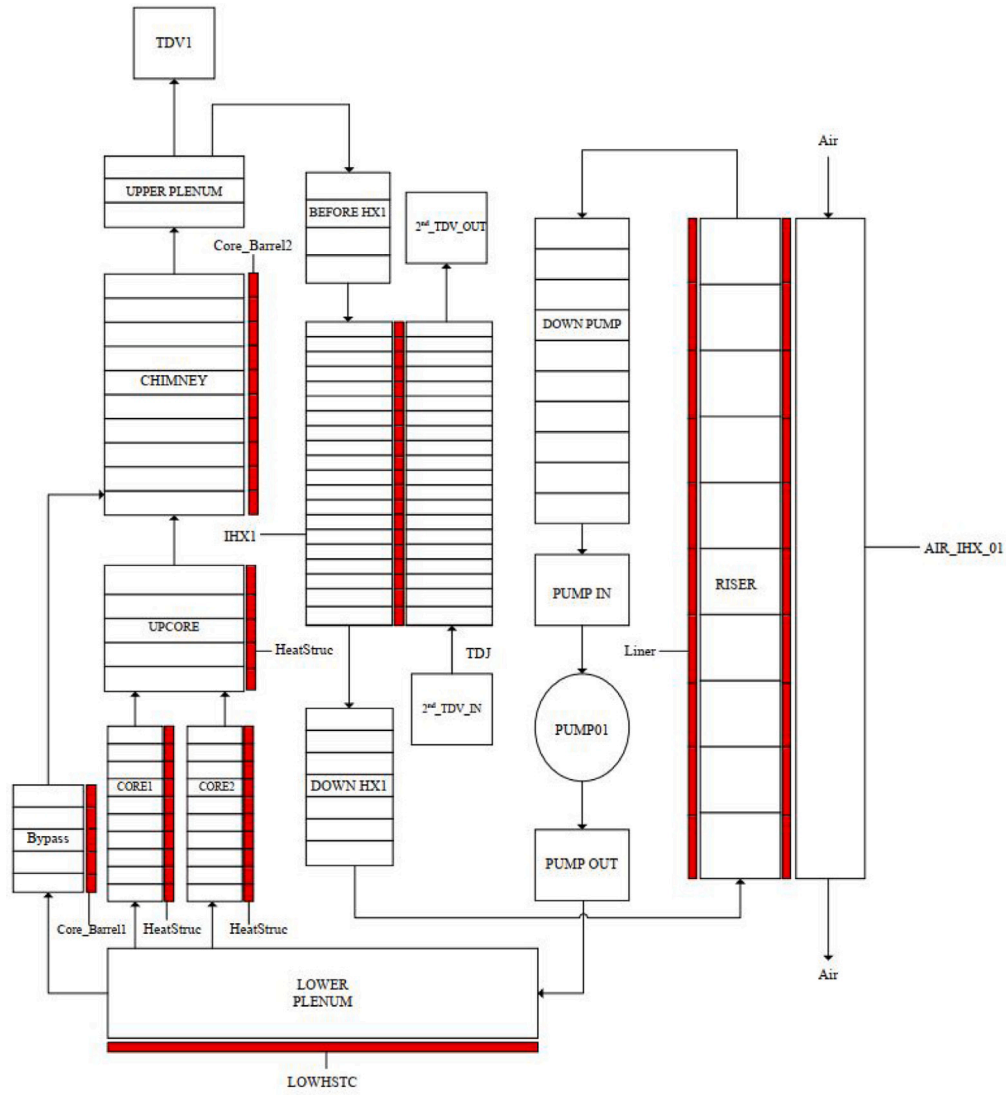


Fig. 3. NUSOL-LMR model of the ABR.

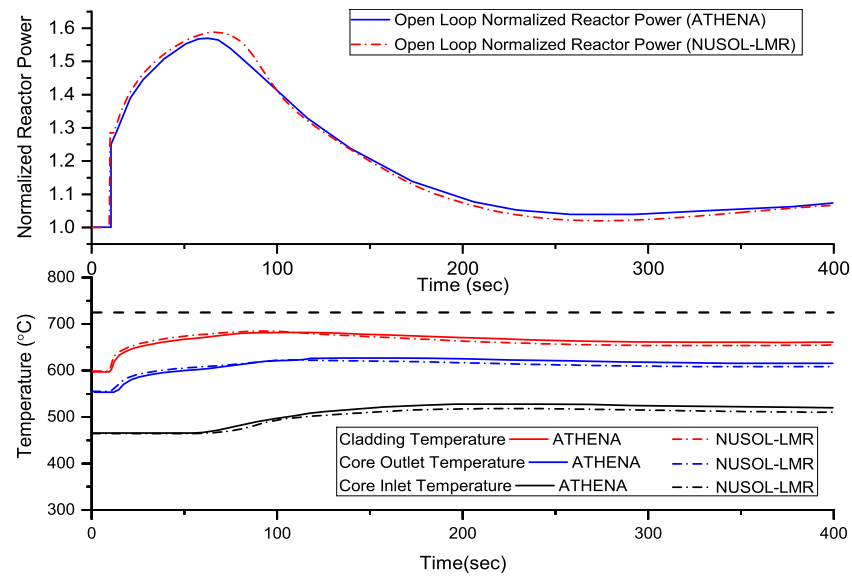


Fig. 4. (a) Comparison of reactor power response on step reactivity insertion of 0.2\$ (b) Comparison of cladding, core outlet, and core inlet temperatures on reactivity insertion of 0.2\$.

Table 2
Geometry of the ABR (Mac Donald et al., 2001).

Parameter	Value
Core	
Fuel Outer Diameter (mm)	5.48
Cladding inner diameter (mm)	6.32
Cladding outer diameter (mm)	7.52
Fuel rod pitch-to-diameter ratio	1.3
Heated length (m)	1.3
Gas plenum length (m)	2.47
Number of fuel assemblies	157
Steam Generator	
Lattice	Triangular
Number of tubes	133 316
Tube pitch-to-diameter ratio	1.2
Tube inner diameter (mm)	13.9
Tube thickness (mm)	4.05
Tube length (m)	6.9
Miscellaneous	
Chimney length (m)	10.5
Core barrel inner diameter (m)	3.16
Core barrel thickness (m)	0.02
Vessel liner inner diameter (m)	5.26
Vessel liner thickness (m)	0.01
Gap between liner and reactor vessel (m)	0.16
Cover gas volume (m3)	59
Pump diameter (m)	0.65
Reactor vessel	
Inner diameter (m)	5.6
Thickness (m)	0.05
Vessel length (m)	18.8

Table 3
Details of the NUSOL-LMR model of the ABR.

Sr.No.	Component name	Component type in code	No of control volumes
1	CORE1	CORE	10
2	CORE2	CORE	10
3	UPCORE	CORE	05
4	CHIMNEY	PIPE	10
5	UPPER PLENUM	PIPE	03
6	TDV1	TDVL	01
7	BEFORE HX1	PIPE	04
8	IHX1(Primary side)	PIPE	20
	IHX1(Secondary side)	PIPE	20
9	DOWN HX1	PIPE	07
10	AIR-IHX-01(RISER)	PIPE	10
11	DOWN PUMP	PIPE	20
12	PUMP IN	PIPE	01
13	PUMP01	PUMP	01
14	PUMP OUT	PIPE	01
15	LOWER PLENUM	PIPE	01
16	BYPASS	PIPE	04

the comparison between the simulated results of ATHENA and NUSOL-LMR codes during the transient of step reactivity insertion of 0.2\$. It is evident from the simulation results that the maximum cladding temperature is well below the safety limit of 725 °C. The results, both in the steady-state and transient-state attained from the NUSOL-LMR code coincide with the results obtained from the ATHENA code. Hence, the model of the ABR established in the NUSOL-LMR code is validated.

4. The ABR reactor power control system

After the model validation, the power regulation control strategy for ABR is devised to evaluate the performance of the newly developed control blocks in the code. The NUSOL-LMR code computes the core power using the point reactor neutron kinetic model and uses the high-order endpoint floating approach to solve the point reactor dynamics equation. The following is the fundamental point reactor

kinetic equation for six delayed neutron precursor groups:

$$\frac{dn}{dt} = \frac{\rho_t - \beta}{\Lambda} . n + \sum_{i=1}^6 \lambda_i c_i \quad (5)$$

$$\frac{dc_i}{dt} = \frac{\beta_i}{\Lambda} . n - \lambda_i c_i \quad (6)$$

According to the point reactor neutron kinetic equation, the total reactivity (ρ_t) will directly affect the neutron flux density, thus affecting the core power. During the reactor's operation, the total reactivity consists of two parts: externally introduced reactivity and feedback reactivity. The externally introduced reactivity includes reactivity changes caused by the automatic movement of control rods (ρ_r) and step reactivity disturbance (ρ_e), the latter being used as a transient control input in the code. Reactivity calculations can be expressed as:

$$\rho_t = \rho_e + \rho_r + \rho_f \quad (7)$$

The NUSOL-LMR code has used the method of lumped parameters to calculate the feedback reactivity (ρ_f). The expression for feedback reactivity is given as:

$$\rho_f = \alpha_f(T_f - T_{f0}) + \alpha_c(T_c - T_{c0}) + \alpha_d(\rho_c - \rho_{c0}) \quad (8)$$

The average values of fuel temperature, coolant temperature, and coolant density have been used in the above expression. In addition, the cumulative effects of fuel thermal expansion and the Doppler effect are expressed by the fuel temperature feedback coefficient (α_f). The consequences of the lower and upper core plates' radial expansion are expressed by the coolant temperature feedback coefficient (α_c). Whereas, for modest changes in density, the coolant density feedback coefficient (α_d) equates to a slightly positive coolant void worth.

$$\rho_r = G_r Z_r \quad (9)$$

The reactivity (ρ_r) introduced by controlling the speed of control rods' movement is given by the expression in Eq. (9). There are 49 control rods in the ABR reactor core, whereas the maximum control rod worth of a single control rod is 0.2\$. The values of β_i , λ_i , Λ , and all the reactivity feedback coefficients are referred from the literature (Davis, 2003; Mac Donald et al., 2001).

4.1. Control strategy for the ABR

The control strategy devised for the power regulation of ABR consists of independent power and temperature control blocks. The primary goal of the power controller in the power control block is to retain the output power at the level programmed into the system. This is accomplished by controlling the reactivity, ' ρ_r ' (by adjusting the speed of control rods), which functions as a controlled input to the point reactor kinetic model of the ABR.

Additionally, in order to reduce the coupling effects between the LBE and reactor power, the temperature control block is used to control the LBE temperature at the core intake. The temperature controller in the control block will regulate the mass flow rate of the coolant in the secondary circuit in order to maintain the core inlet temperature of the LBE close to its reference value (steady-state value). Fig. 5 shows the block diagram of the designed control strategy of the ABR.

During any transient scenario, both control blocks will work together to attain the objective of keeping the anticipated closed-loop power level within $\pm 2\%$ of the error band (Ponciroli et al., 2015; Shen et al., 2023). Furthermore, the power and temperature controllers will track their reference values with reduced overshoot, shorter settling time, and minimal steady-state error.

Table 4
Steady-state results of the ABR.

Parameter	NUSOL-LMR	ATHENA	DOSTAL ET AL (Dostal et al., 2002)
Primary coolant system			
Core power (MW)	700	700	700
Core inlet temperature (°C)	464.26	465.9	467
Core outlet temperature (°C)	555.27	554.3	554.9
Primary coolant mass flow rate (kg/s)	54 400	54 420	54 420
Secondary coolant system			
Pressure(MPa)	15	15	15
Feedwater flow rate (kg/s)	333.3	333.3	316.7
Feedwater temperature (°C)	280	280	280

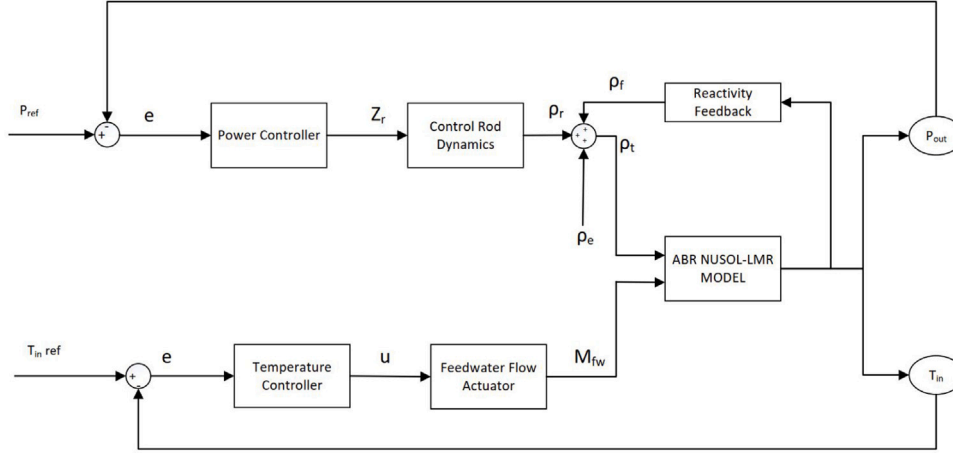


Fig. 5. Block diagram of the control strategy of the ABR.

4.1.1. Power control block

The closed-loop power control block consists of the power controller, control rod dynamics block, and ABR NUSOL-LMR model block. The input signal to the power controller is the error signal (deviation between reference and actual power), whereas the power controller's output is a signal representing the speed of the control rod (Z_r). This speed signal is supplied to the control rod dynamics block, which transforms it into reactivity, ' ρ_r ' (by using Eq. (9)). Eventually, the ABR NUSOL-LMR model block, which generates the reactor power, uses this reactivity (ρ_r) as an input signal. In addition to the feedwater power control loop, the power control block has an internal feedback loop for reactivity feedback (ρ_f). The power control block in the NUSOL-LMR code is equipped with an option to use two different controllers:

- Proportional, Integral and Derivative Controller (PID)

The power control block when used with the PID controller, its mathematical equation in the time domain is given as:

$$Z_r = k_p e + k_i \int e dt + k_d \frac{de}{dt} \quad (10)$$

- Sliding Mode Controller with a PI Sliding Surface (PISM)

While designing the PISM controller, it is assumed that all the system states are measurable (Hui and Yuan, 2020). Moreover, the performance of the SMC depends on the construction of an appropriate sliding surface ($s(t)$). Therefore, in this regard, PI and SMC controllers are integrated and employed together for the construction of sliding surface. The closed-loop power control block diagram of the ABR with the PISM controller is illustrated in Fig. 6.

The mathematical expression for the PI sliding surface and its time derivative is as follows:

$$s = k_1 e + k_2 \int e dt \quad (11)$$

$$\dot{s} = k_1 \dot{e} + k_2 e \quad (12)$$

The error dynamics are given as:

$$e = P_{ref} - P_{out} \quad (13)$$

$$\dot{e} = \dot{P}_{ref} - \dot{P}_{out} \quad (14)$$

Putting the values of Eq. (14) into (12),

$$\dot{s} = k_1 (\dot{P}_{ref} - \dot{P}_{out}) + k_2 e \quad (15)$$

In order to calculate the control law of the PISM controller, the general form of the six-group point reactor kinetic equation presented in (5) and (6) is converted into the normalized form (Hui and Yuan, 2020):

$$\begin{aligned} \dot{P}_{out} &= \frac{dn_r}{dt} = \frac{\rho_t - \beta}{\Lambda} n_r + \sum_{i=1}^6 \frac{\beta_i}{\Lambda} c_{ri} \\ \frac{dc_r}{dt} &= \lambda_i n_r - \lambda_i c_{ri} \end{aligned} \quad (16)$$

Substituting the value of $\dot{P}_{out}$ into Eq. (15) gives:

$$k_1 \left(\dot{P}_{ref} - \left(\frac{\rho_t - \beta}{\Lambda} n_r + \sum_{i=1}^6 \frac{\beta_i}{\Lambda} c_{ri} \right) \right) + k_2 e = \dot{s} \quad (17)$$

From Gao and Hung (1993), Luo et al. (2022), the reaching law dynamics are:

$$\dot{s} = -\omega \text{Sign}(s) - k_3 s \quad (18)$$

In order to reduce the chattering phenomenon, the 'Sigmoid function' will be employed in place of the 'Sign function', and the reaching law presented in Eq. (18) is modified as (Wang et al., 2023):

$$\dot{s} = -\omega \text{Sigmoid}(s) - k_3 s \quad (19)$$

where,

$$\text{Sigmoid}(s) = \frac{s}{|s| + \xi}$$

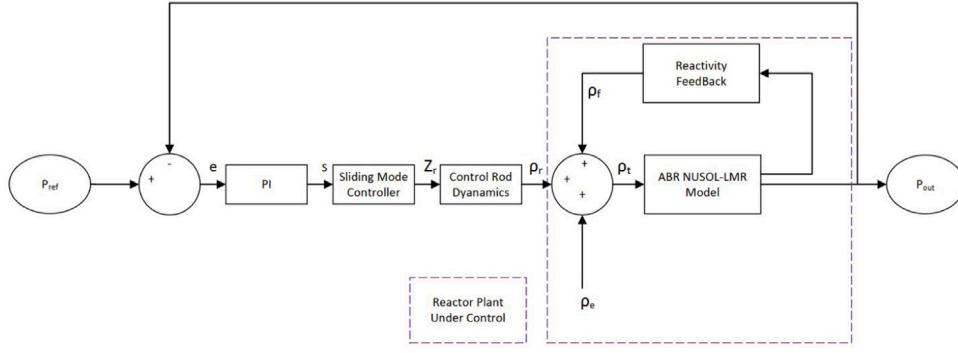


Fig. 6. The power control block diagram of the ABR with the PISM controller.

' ξ ' is a small positive constant, Eq. (17) can be re-written as:

$$k_1 \left(\dot{P}_{ref} - \left(\frac{\rho_t - \beta}{\Lambda} n_r + \sum_{i=1}^6 \frac{\beta_i}{\Lambda} c_{ri} \right) \right) + k_2 e = -\omega \text{Sigmoid}(s) - k_3 s \quad (20)$$

Substituting the values of ' ρ_t ' from Eq. (7) yields:

$$\rho_r = \frac{\Lambda \dot{P}_{ref}}{n_r} + \Lambda \frac{k_2}{k_1} \frac{e}{n_r} + \beta - \frac{1}{n_r} \sum_{i=1}^6 \beta_i c_{ri} + \frac{\Lambda \omega}{k_1 n_r} \text{Sigmoid}(s) + \frac{\Lambda k_3}{k_1 n_r} s - \rho_e - \rho_f \quad (21)$$

From Eq. (9),

$$\rho_r = G_r \int Z_r$$

Finally, the PISM control law, in terms of control rod speed (Z_r), is given as:

$$G_r \int Z_r = \frac{\Lambda \dot{P}_{ref}}{n_r} + \Lambda \frac{k_2}{k_1} \frac{e}{n_r} + \beta - \frac{1}{n_r} \sum_{i=1}^6 \beta_i c_{ri} - \rho_e - \rho_f + \frac{\omega \Lambda}{k_1 n_r} \text{Sigmoid}(s) + \frac{k_3 \Lambda}{k_1 n_r} s \quad (22)$$

$$Z_r = \frac{1}{G_r} \left(\frac{d}{dt} \left(\frac{\Lambda \dot{P}_{ref}}{n_r} + \Lambda \frac{k_2}{k_1} \frac{e}{n_r} + \beta - \frac{1}{n_r} \sum_{i=1}^6 \beta_i c_{ri} - \rho_e - \rho_f + \frac{\omega \Lambda}{k_1 n_r} \text{Sigmoid}(s) + \frac{k_3 \Lambda}{k_1 n_r} s \right) \right)$$

The PISM control law, derived in Eq. (22), is the sum of the equivalent control law (Z_{req}) and switching control law (Z_{rsu}).

$$Z_r = Z_{req} + Z_{rsu} \quad (23)$$

where,

$$Z_{req} = \frac{1}{G_r} \left(\frac{d}{dt} \left(\frac{\Lambda \dot{P}_{ref}}{n_r} + \Lambda \frac{k_2}{k_1} \frac{e}{n_r} + \beta - \frac{1}{n_r} \sum_{i=1}^6 \beta_i c_{ri} - \rho_e - \rho_f \right) \right)$$

$$Z_{rsu} = \frac{1}{G_r} \left(\frac{d}{dt} \left(\frac{\omega \Lambda}{k_1 n_r} \text{Sigmoid}(s) + \frac{k_3 \Lambda}{k_1 n_r} s \right) \right)$$

The stability of the derived control law of the PISM controller needs to be ensured; therefore, a positive definite Lyapunov function is taken as:

$$V_{SMC} = \frac{1}{2} s^2 \quad (24)$$

The time derivative of the Lyapunov function gives:

$$\dot{V}_{SMC} = s \dot{s} \quad (25)$$

Using the values of ' s ' and ' $\dot{P}_{out}$ ' from Eqs. (15) and (16),

$$\dot{V}_{SMC} = s (k_1 (\dot{P}_{ref} - \dot{P}_{out}) + k_2 e)$$

$$\dot{V}_{SMC} = s \left(k_1 \left(\dot{P}_{ref} - \left(\frac{\rho_t - \beta}{\Lambda} n_r + \sum_{i=1}^6 \frac{\beta_i}{\Lambda} c_{ri} \right) \right) + k_2 e \right)$$

Now, substituting the value of ' ρ_r ' from Eq. (21) gives:

$$\dot{V}_{SMC} = s (-\omega \text{Sigmoid}(s) - k_3 s) \quad (26)$$

For $\omega > 0$ and $k_3 > 0$ will make $\dot{V}_{smc}$ negative definite. Hence, by choosing the appropriate positive gains of the reaching law given in Eq. (19) along with the positive gains of PI sliding surface, the PISM stability and error convergence will always be ensured. Fig. 7 shows the power control block diagram of the ABR with the derived PISM control law.

4.1.2. Temperature control block

As the SMC control law for feed water temperature control has not been derived yet, the design and implementation of only the PI temperature controller is part of the control strategy's second phase. A temperature controller, an actuator for the feed water flow, and the ABR NUSOL-LMR model block are the components that make up the closed-loop temperature control block. The error signal, which is the disparity between the LBE temperature-programmed value (T_{in} ref) and the actual value measured at the core intake (T_{in}), is sent to the temperature controller. The secondary coolant flow will be adjusted by the feed water flow actuator in response to the temperature controller's outcome. Finally, the ABR NUSOL-LMR model block, based on the flow rate of coolant flowing on the secondary side, will determine the LBE temperature at the core intake. The time-domain mathematical equation for the PI controller is as follows:

$$u = k_p \cdot e + k_i \int e dt \quad (27)$$

Fig. 8 illustrates the temperature control block diagram of the ABR. The feed water flow actuator in the block diagram is represented by TDJ (time-dependent Junction) in the NUSOL-LMR code (as already discussed in Section 2.1). Furthermore, TDJ will control flow inlet velocity of the secondary coolant based on input from the PI controller.

5. Simulation and results

The efficiency of the aforementioned control scheme is assessed using three distinct transient situations. In the first instance, external step reactivity (square wave pattern) is introduced, while in the second instance, an analysis is done on a 2.5% decrease in primary coolant flow brought on by the primary coolant pump transient. Finally, under the intended power level changes to 110%, 80%, and 70% of full power, the developed control strategy's load-following competence is evaluated. The values of PI, PID, and PISM gain parameters during these three distinct transient scenarios are tabulated in Table 5. The gains of a PID controller are chosen through the Ziegler–Nichols step response method. At the same time, the gain parameters of the PISM and PI temperature controller are determined through trial and error. Moreover, after taking into consideration the physical constraints that

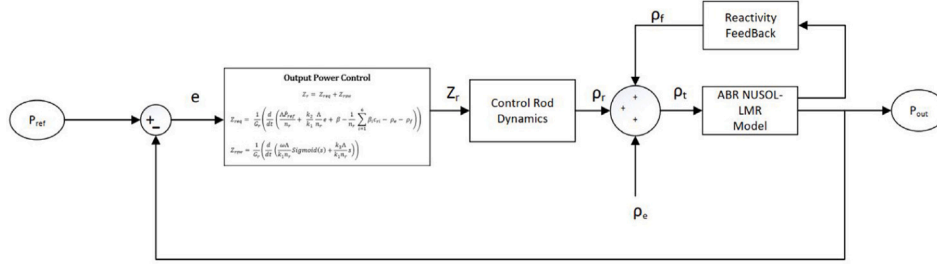


Fig. 7. The power control block diagram of the ABR with the PISM control law.

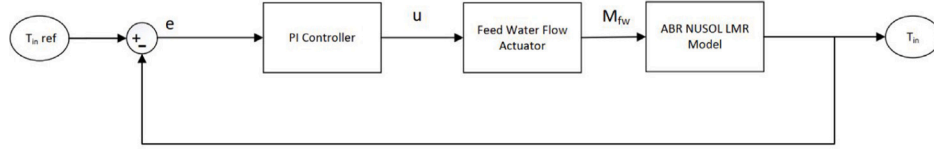


Fig. 8. Temperature control block diagram of the ABR.

Table 5
Controller gain parameter values.

Power controller		Temperature controller
PID	PISM	PI
$k_p = 0.0062, k_i = 0.04, k_d = 0.002$		$k_p = -0.004, k_i = -0.0001$
$k_1 = 0.00598, k_2 = 0.0000225, k_3 = 0.0024, \omega = 18.24$		

are imposed on the control rod speed by practical applications, the control rod speed has been fixed to $|Z_r| \leq 0.018$ m/s during transient events.

5.1. Case 1: Square wave reactivity insertion

In this case, the square wave reactivity insertion scenario is taken into consideration and the pattern of reactivity insertion is shown in Fig. 9 (a). The core is injected with 0.03 \$ of step reactivity at 20 s. Due to this reactivity insertion, the reactor power suddenly enhances from the steady state value (at which the reactor is initially operating). At this stage, the reactor is running in an open-loop without any controller. Therefore, after a brief boost owing to an increase in negative reactivity feedback (a rise in coolant and fuel temperature), the power of the reactor will gradually decline.

Likewise, step reactivity of -0.03 \$ is injected into the core at 500 s, resulting in an initial instantaneous decrease in reactor power; eventually, the power will increase owing to reactivity feedback (a fall in coolant and fuel temperature). The open-loop response of the normalized power level during the transient event is shown in Fig. 9 (b).

A closed-loop control strategy covered in Section 4.1 is now utilized to enhance the ABR's response to this transient event. The closed-loop characteristics of normalized reactor power are shown in Fig. 9 (c), while the speed at which the control rod travels to deal with the consequences of reactivity insertion is shown in Fig. 10. Figs. 11 and 12 draw comparisons between the open and closed-loop core average coolant temperatures and primary coolant inlet temperatures, respectively. The variation of water's flow rate in the secondary circuit due to the temperature controller action has been illustrated in Fig. 13.

The findings illustrate that the ABR's closed-loop performance is superior and more effective than that of an open-loop performance. The ABR's closed-loop response further demonstrates that the power control block comprising a PISM controller shows better power regulation capability than that of a simple PID controller when used simultaneously with a PI temperature controller.

Moreover, the simulation results also elaborate that using the PISM controller with the reaching law containing a sign function (Eq. (18))

will cause chattering in the ABR's closed-loop reactor power response and core average coolant temperature. This chattering is caused by the high-frequency switching control signal of the PISM controller for keeping the trajectory of the closed-loop system on the sliding surface (Fig. 10 (c)). As a consequence, it may result in excessive wear and tear on moving mechanical parts and extreme heat losses in the actuator's electrical power circuit. Therefore, to avoid this chattering, reaching law with the sigmoid function (Eq. (19)) is employed for smoothing out the discontinuity in the control signal (Wang et al., 2013; Huang et al., 2012).

5.2. Case 2: Primary coolant flow transient

The step change in the LBE flow rate caused by the primary coolant pump malfunction is examined in the second scenario. Fig. 14 compares the open and closed-loop reactor power responses of the ABR. Prior to the initiation of the transient event, the reactor was operating normally at 100% full power. At 20 s, due to the primary pump fault, the primary coolant flow is reduced by 2.5% of its steady-state value. The sudden change in primary coolant flow will cause the reactor power, core average temperature, and primary coolant inlet temperature to change. ABR inherent safety features (negative feedback coefficients) during the open-loop will make sure that reactor safety is not compromised under this transient event.

The ABR's open-loop normalized reactor power response shows higher values of settling time, larger overshoot, and greater steady-state error. In addition, without a temperature controller, the primary coolant inlet temperature also deviates from its steady-state value.

However, by employing the devised control strategy, the simulation results shown in Figs. 14–16 clearly indicate that the performance of the ABR under the closed-loop is better. The power control block makes sure that the normalized reactor power reaches its steady-state value in the least amount of time with reduced overshoot and minimal steady-state error. Furthermore, the temperature control block controls the water's flow rate to keep the LBE temperature at core intake near to its reference value. Table 6 summarizes the performance parameters of the power controllers in the transient circumstances stated above.

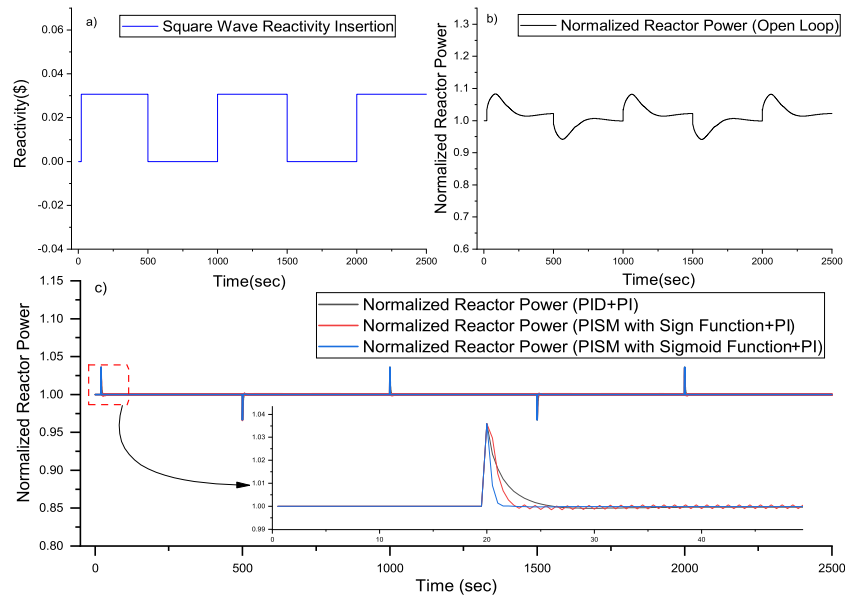


Fig. 9. (a) Square wave reactivity insertion transient (b) Open-loop normalized reactor power response during the square wave reactivity insertion transient (c) Closed-Loop normalized reactor power response during the square wave reactivity insertion transient.

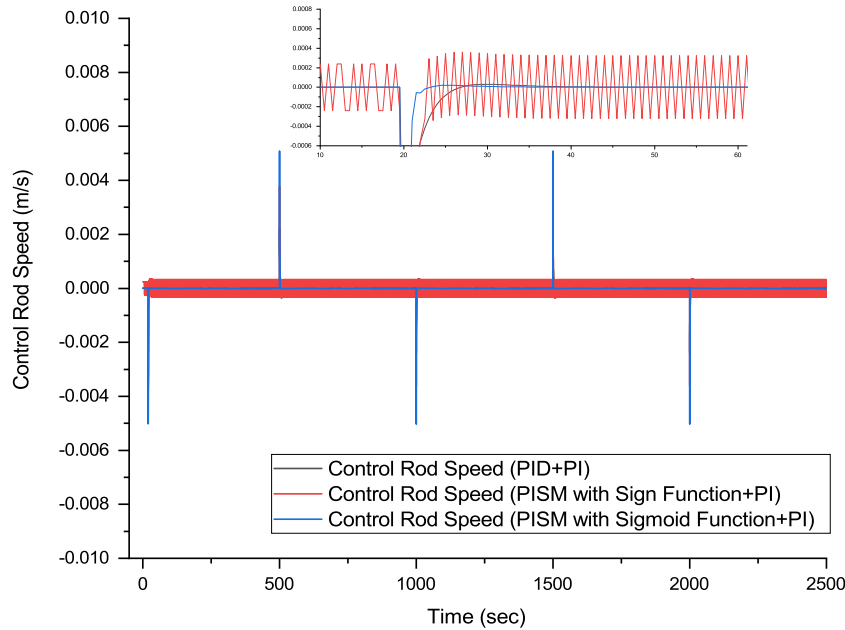


Fig. 10. Control rod speed during the square wave reactivity insertion transient.

Table 6

Performance parameters of power controllers during the reactivity insertion and the flow transient.

Transient event	Power controller		Open loop
	PID	PISM	
	Overshoot (%)		Overshoot (%)
Reactivity insertion	3.7	3.43	8.2
Flow transient	0.27	0.169	1.1

5.3. Case 3: Change in demand power level

The third and final scenario evaluates the proposed control scheme's load-following capabilities. The power and temperature controllers

designed as part of the control approach will both operate concurrently to improve the ABR's closed-loop performance. Initially, the power is boosted to 110% of full power, and subsequently decreased to 80% and 70% of full power, respectively.

The hypothetical scenario of a power increase to 110% of full power at 20 s is illustrated in Fig. 17. During this transient event, it is assumed that the safety system is not actuated. As a result, the PID, PISM with sign function, and PISM with sigmoid function in the power control block have tracked the desired power level with an overshoot of 0.7%, 0.62%, and 0.51%, respectively. In less than 10 s, the closed-loop normalized reactor power settles within the $\pm 2\%$ error band for all the controllers. The simulation outcomes also demonstrate that the chattering is caused when the PISM controller uses the sign function in its reaching law. Therefore, this chattering is avoided by employing the

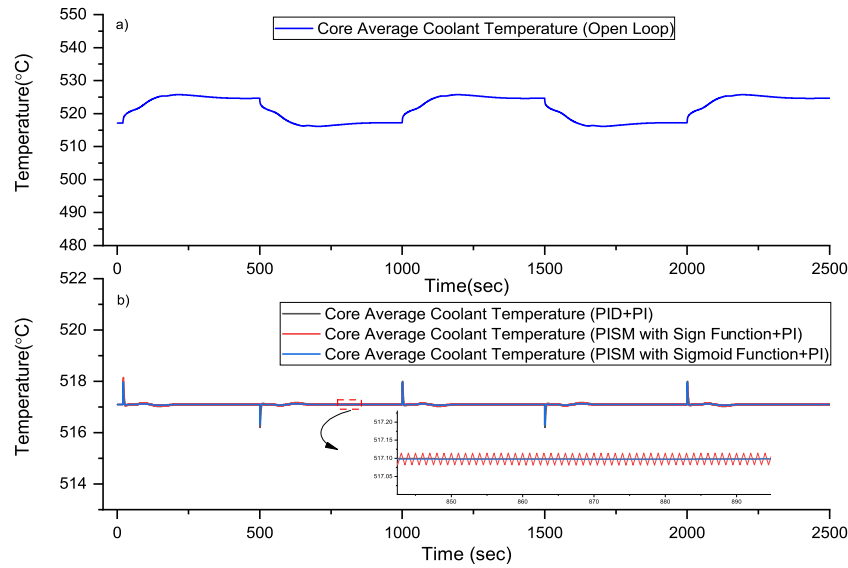


Fig. 11. Core average coolant temperature during the square wave reactivity insertion transient.

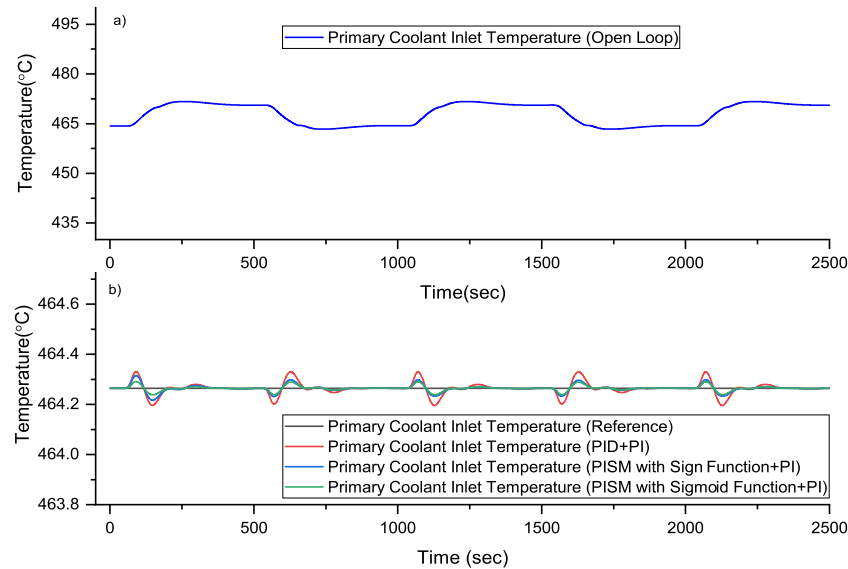


Fig. 12. Primary coolant inlet temperature during the square wave reactivity insertion transient.

PISM controller with the sigmoid function (Wang et al., 2013; Huang et al., 2012).

The total reactivity in the core due to the combined effect of feedback reactivity and control rod movement has been illustrated in Fig. 18. At the same time, the variation in core average coolant temperature and primary coolant inlet temperature during the initiation of the transient event has been depicted in Fig. 19. Along with the power controller, the PI temperature controller has also tracked the primary coolant inlet temperature to its reference value with an overshoot of 1.01% and a settling time of 121 s.

Moreover, when the desired power level is reduced to 80% of full power at 20 s, the simulation results shown in Fig. 20 indicates that the PID and PISM controllers in the power control block, efficiently track the change in power level with an overshoot of 2.3% and 1.7%, respectively. The PID controller took 11.9 s, whereas the PISM responded in less than 10 s to settle the reactor power within the designated power error band. The variation of total reactivity inside the core during the transient event has been shown by Fig. 21. Fig. 22 (a) illustrates that the PI temperature controller effectively controls the LBE inlet temperature

with an overshoot of 1.8% and a settling time of 128 s. Whereas Fig. 22 (b) shows the profile of coolant average temperature within the core.

Lastly, the power is reduced to 70% of full power, the ABR's power regulation system continues to operate precisely, and the power controllers (PID and PISM) regulate and track the reactor power with an overshoot of 4.2% and 3.2%, respectively (as illustrated in Fig. 23). The PID controller takes nearly 17.4 s to settle the reactor power within the designated power error band. The response of PISM is quite swift, and it makes the reactor power to settle within the error band of $\pm 2\%$ in less than 10 s. Fig. 24 shows the trend of total reactivity inside the core during the demand power change to 70% of full power. Furthermore, Figs. 25 (a) and (b) show the primary coolant inlet temperature and core average coolant temperature, respectively. In addition, the temperature control block controls the primary coolant inlet temperature with an overshoot of 2.8% and a settling time of 139 s while operating together with the power control block. Table 7 provides an overview of the performance parameters of the controllers during change in demand power level transient scenarios. The response of the PID and PISM controllers in the presence of a bounded measurement noise during

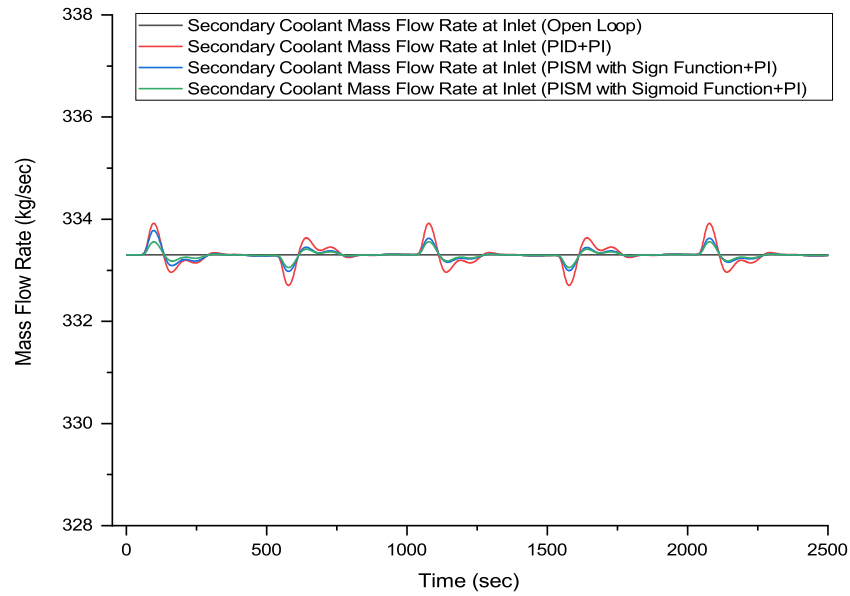


Fig. 13. Secondary coolant mass flow rate during the square wave reactivity insertion transient.

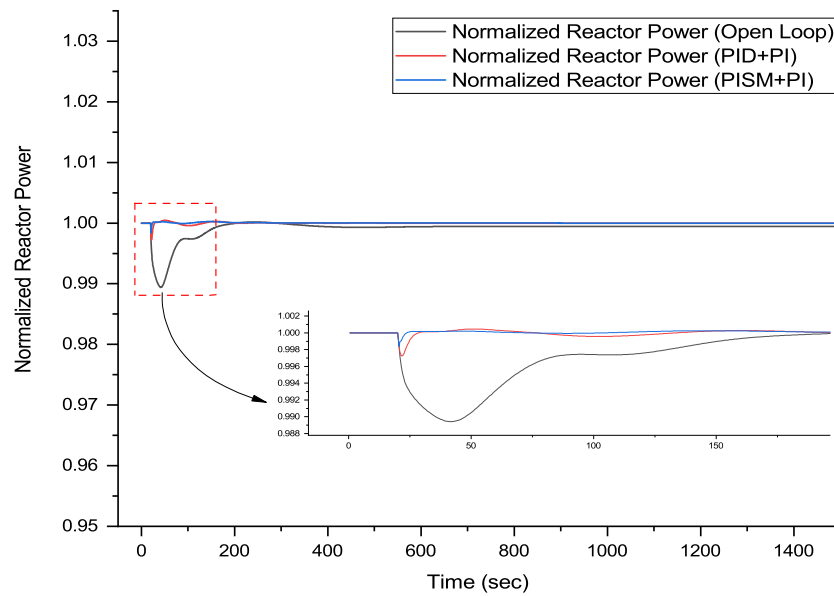


Fig. 14. Normalized reactor power response during the flow transient.

the power reduction to 70% of full power is depicted in Fig. 26. The bounded measurement noise signal has mean and standard deviation values of 0 and 0.00294, respectively. Both the power controllers responded effectively and efficiently, maintaining the reactor's power within an error band of $\pm 2\%$. It is evident from the simulation results that the PISM controller regulated the reactor power with reduced overshoot and shorter settling time than the PID controller, even in the presence of bounded measurement noise.

6. Conclusion

Control blocks consisting of conventional PI, PID, and SMC controllers are developed and implemented in the NUSOL-LMR code. The modeled control blocks enhanced the capability of the code to perform the closed-loop analysis of metal-cooled fast reactors under the controller action during transient events. Furthermore, for evaluating

the performance of the developed control blocks, a control strategy for reactor power control of the ABR is presented in this paper.

Due to the inherent design characteristics and features of LFRs, their power control schemes are different from those of PWRs. The power control scheme of PWRs uses a single controller to control the movement of control rods upon receiving error signals from the power mismatch and coolant average temperature (T_{avg}) channels. In contrast, the power control schemes of the LFRs (M2LFR, SNCLFR, etc.) discussed in the literature, and the designed power control strategy of the ABR in this research work have utilized simultaneously working independent power and temperature controllers.

Using the NUSOL-LMR code, the nodalization scheme of the ABR was constructed in the first phase to investigate and assess the thermal-hydraulic performance of the said model. Following the model's validation, the designed control strategy for the ABR has been evaluated

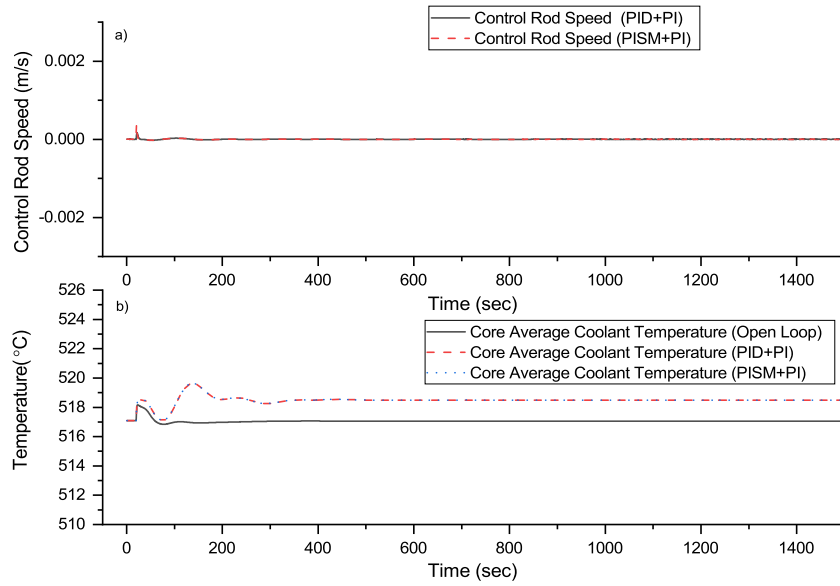


Fig. 15. (a) Control rod speed during the flow transient (b) Core average coolant temperature during the flow transient.

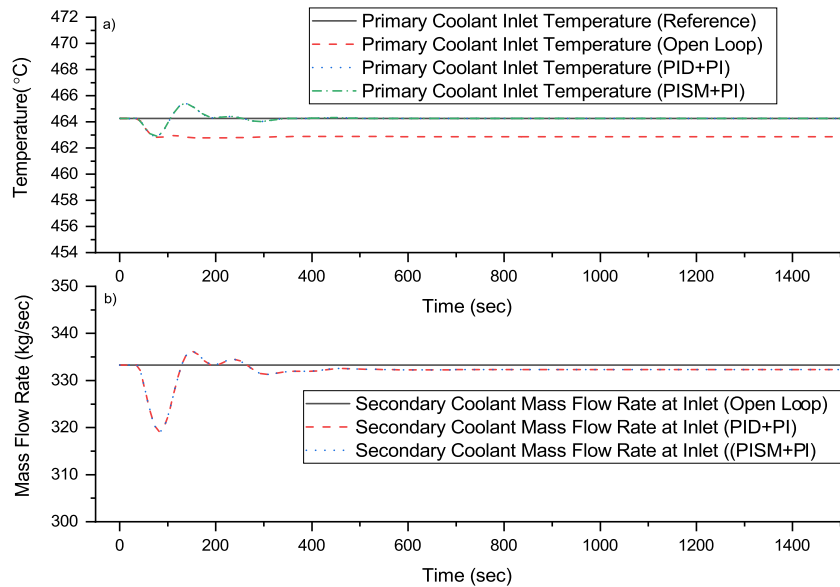


Fig. 16. (a) Primary coolant inlet temperature during the flow transient (b) Secondary coolant mass flow rate during the flow transient.

Table 7

Performance parameters of the controllers during the demand power level change transient.

Transient event	Power controller				Temperature controller	
	PID		PISM		PI	
	Overshoot (%)	Settling time (s)	Overshoot (%)	Response time (s)	Overshoot (%)	Settling time (s)
110% of full power	0.7	<10	0.51	<10	1.01	121
80% of full power	2.3	<15	1.7	<10	1.8	128
70% of full power	4.2	<20	3.2	<10	2.8	139

for three distinct transient situations. The simulation results demonstrate that implementing the closed-loop control scheme eliminates the shortcomings of the open-loop response to reactivity insertion and flow transients (more significant overshoot, greater settling time, and larger steady-state error).

Similarly, in the last case of the load-following condition, the temperature controller working in collaboration with the power controller has ensured that the reactor output power reaches the error band of $\pm 2\%$ with shorter settling time, reduced overshoot, and minimal steady-state error. It is evident from the closed-loop simulation results that

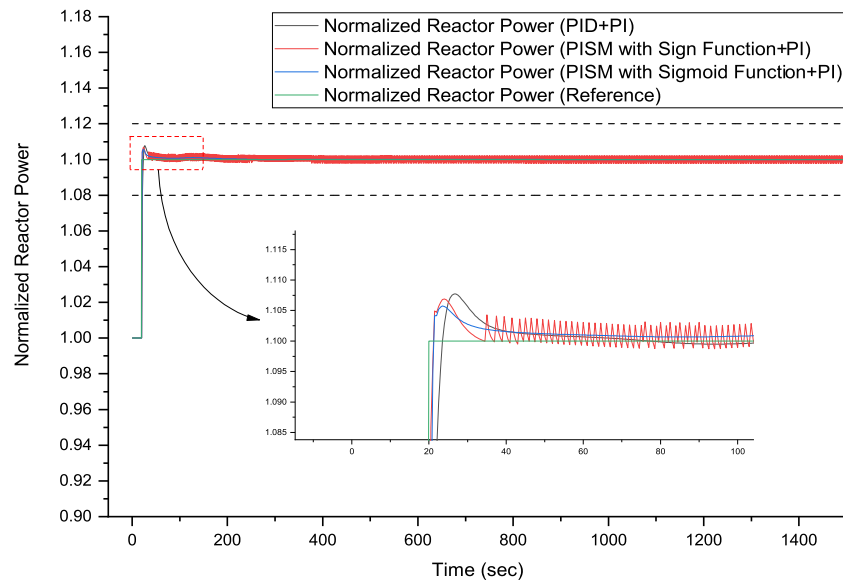


Fig. 17. Normalized reactor power response during the transient of demand power level change to 110% of full power.

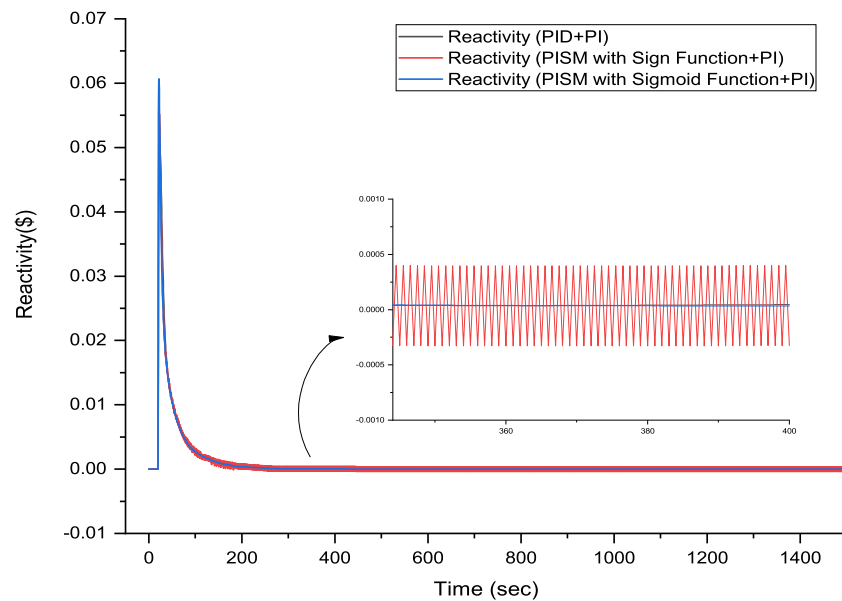


Fig. 18. Total reactivity introduced in the core during the transient of demand power level change to 110% of full power.

the PISM controller response in the case of transient is superior to the conventional PID controller in the power control block. This superiority is reflected in the rapid and accurate tracking of the output power to the desired reference value, with the tracking error approaching zero asymptotically. In addition, the Lyapunov stability theorem is employed to ensure the stability of the control law implemented in the PISM controller. This theorem certifies that the non-linear reactor's overall dynamics remain stable throughout the reaching and sliding phases.

Hence, with the implementation of control blocks in the NUSOL-LMR code, the closed-loop performance of the ABR during transient events is quite satisfactory. Furthermore, an observer-based sliding mode controller using NUSOL-LMR will be developed in the future. In addition, after developing more component models in the NUSOL-LMR code, the developed control blocks will be utilized for the closed-loop analysis of the complete power conversion cycle of the LFRs.

CRediT authorship contribution statement

Mujtaba Mujahid: Methodology, Software, Validation, Writing – original draft, Conceptualization. **Wang Jingxin:** Validation, Visualization. **Ghulam Jillani:** Investigation, Writing – review & editing. **Kun Fan:** Data curation, Resources. **Jianqiang Shan:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

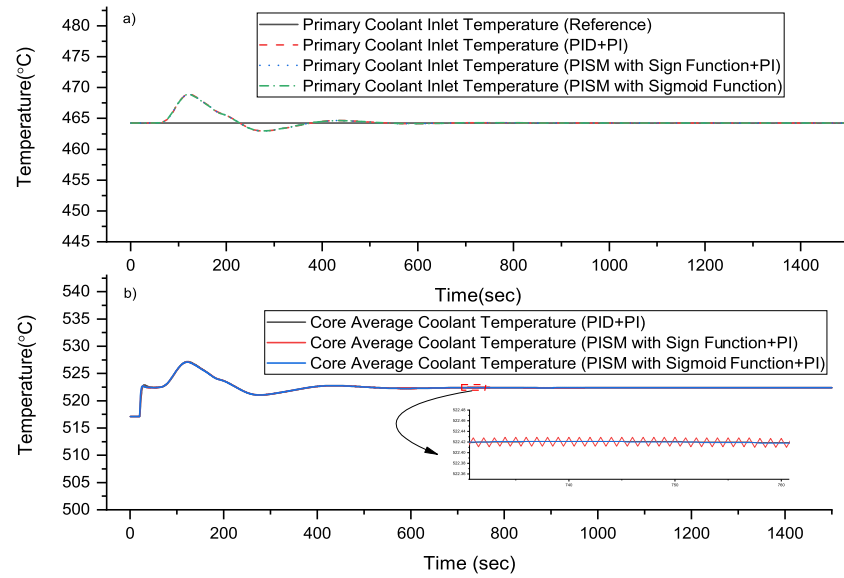


Fig. 19. (a) Primary coolant inlet temperature during the transient of demand power level change to 110% of full power (b) Core average coolant temperature during the transient of demand power level change to 110% of full power.

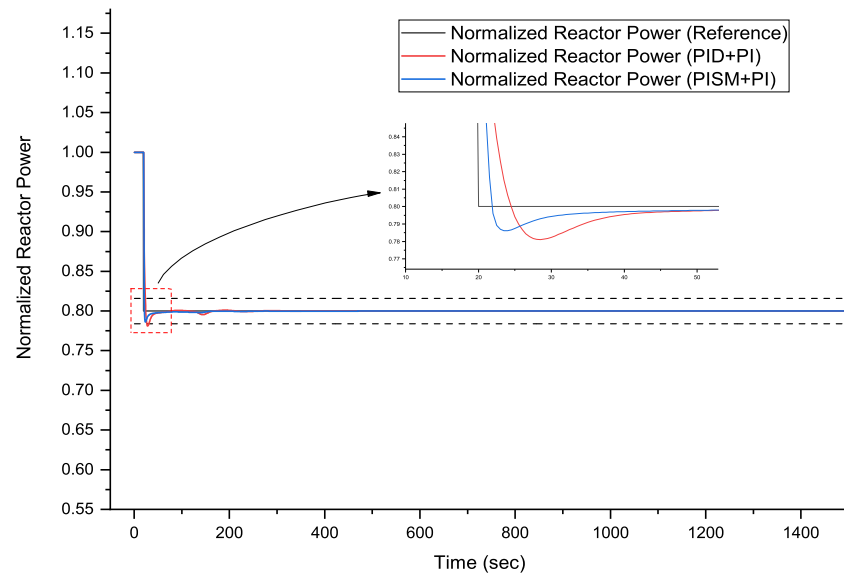


Fig. 20. Normalized reactor power response during the transient of demand power level change to 80% of full power.

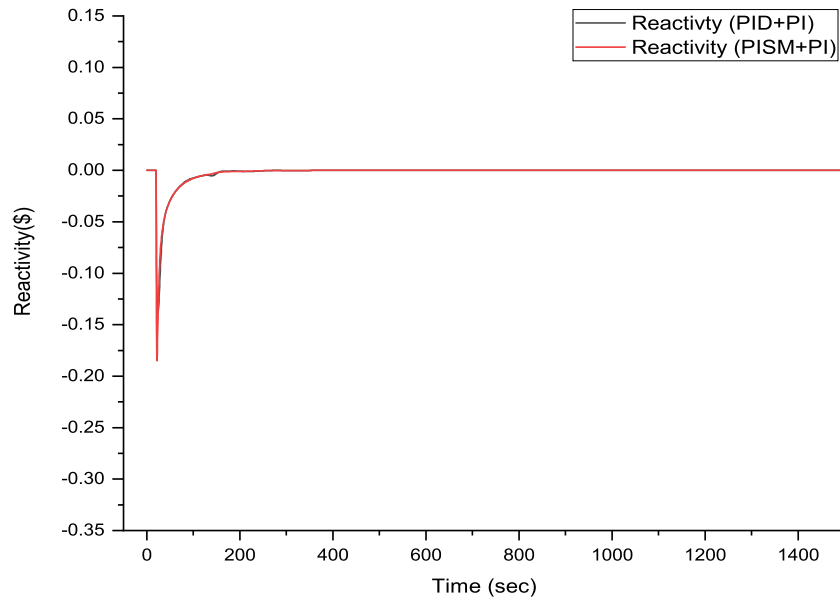


Fig. 21. Total reactivity introduced in the core during the transient of demand power level change to 80% of full power.

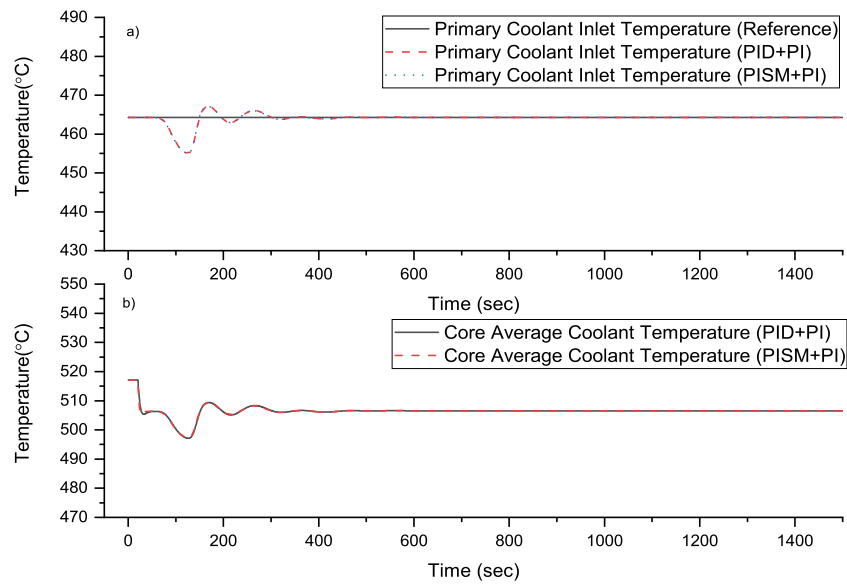


Fig. 22. (a) Primary coolant inlet temperature during the transient of demand power level change to 80% of full power (b) Core average coolant temperature during the transient of demand power level change to 80% of full power.

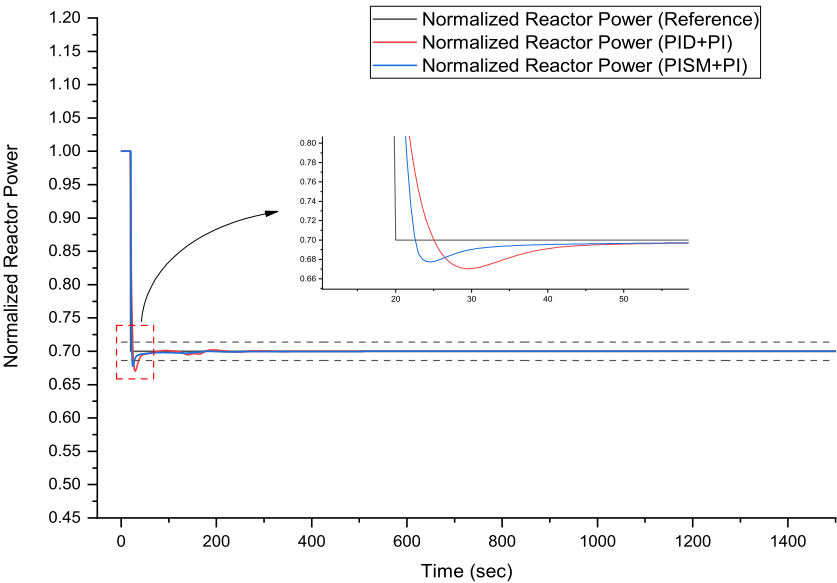


Fig. 23. Normalized reactor power response during the transient of demand power level change to 70% of full power.

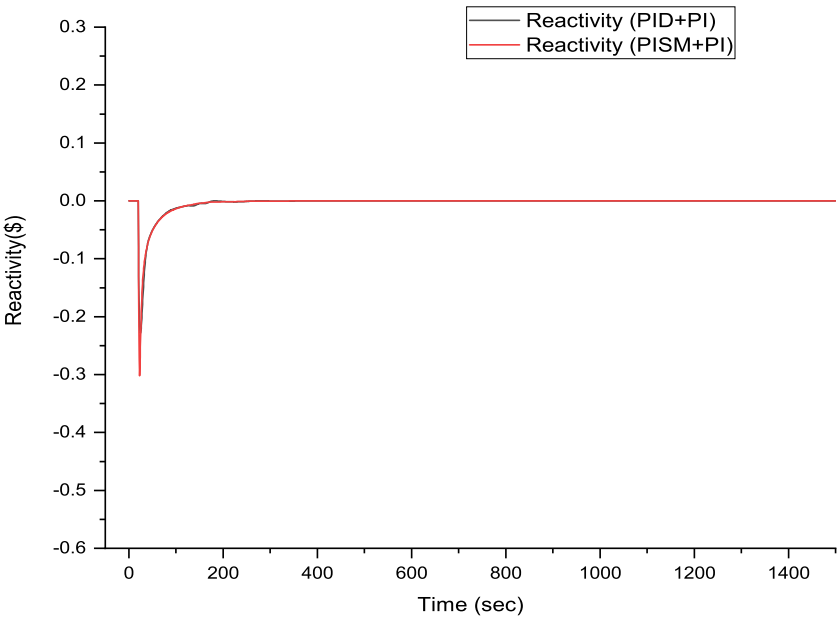


Fig. 24. Total reactivity introduced in the core during the transient of demand power level change to 70% of full power.

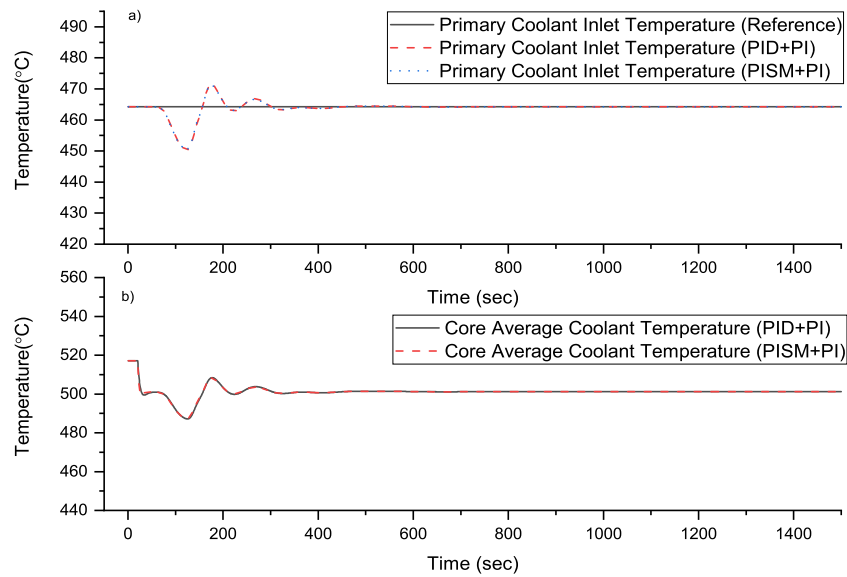


Fig. 25. (a) Primary coolant inlet temperature during the transient of demand power level change to 70% of full power (b) Core average coolant temperature during the transient of demand power level change to 70% of full power.

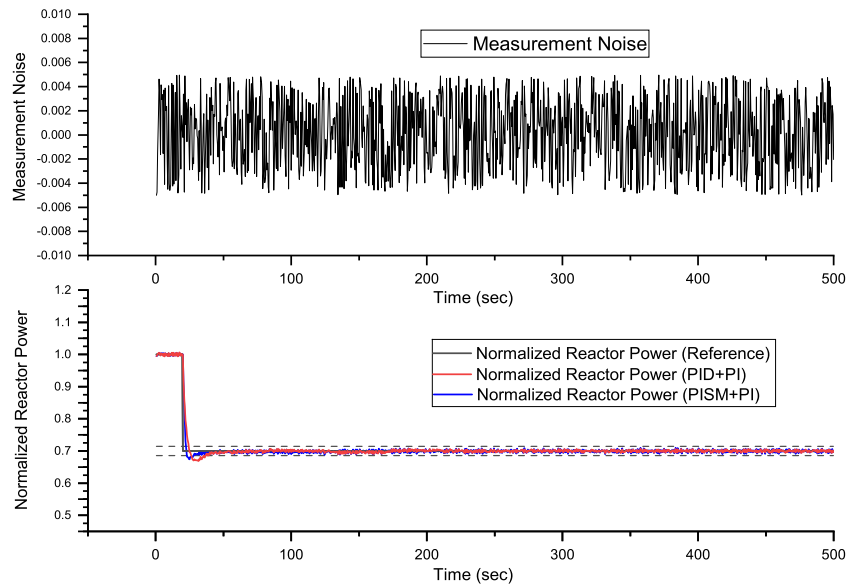


Fig. 26. Normalized reactor power response during the transient of demand power level change to 70% of full power in the presence of measurement noise.

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